Prompt 2 - trial 3 (seed 1240) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a Deep Neural Network to on compounds active against the D(2) dopamine receptor (UniProt: P14416) vs inactive, evaluate performance, compare Molanchor vs SHAP and MolCE vs Counterfactuals on the same correctly predicted active compound. Summarize agreements and disagreements.

Assistant

🍼 Let me start by finding a relevant dataset for the D(2) dopamine receptor and then train a Deep Neural Network model!

(Calling tool `find_datasets`)

Assistant

🎯 Found the D(2) dopamine receptor dataset! Let me load it and set up the training pipeline.

(Calling tool `load_dataset`)

Assistant

📊 Dataset loaded with 8,348 compounds. Now let me compute ECFP fingerprints and split the data.

(Calling tool `compute_features`)

Assistant

✨ Features computed! Now splitting the dataset for training.

(Calling tool `split_dataset`)

Assistant

📊 Dataset split successfully! Now training the Deep Neural Network model. This may take a moment...

(Calling tool `train_model`)

Assistant

⏳ DNN model training in progress. Let me check the status...

(Calling tool `check_training`)

Assistant

⏳ Still training... Let me check again in a moment.

(Calling tool `check_training`)

Assistant

⏳ Training continues...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Still training...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

⏳ Training in progress...

(Calling tool `check_training`)

Assistant

🎉 **DNN Model Training Complete!** Excellent performance achieved:

Metric	Value
Accuracy	98.64%
AUC	0.9957
Balanced Accuracy	0.9864
F1 Score	0.9864
Precision	0.9849
Recall	0.9880

Now let me export predictions and select a correctly predicted active compound for explainability analysis.

(Calling tool `export_predictions`)

Assistant

📊 Predictions exported! Now let me select a correctly predicted active compound for XAI analysis.

(Calling tool `select_compound_for_xai`)

Assistant

🎯 **Selected Compound for XAI:**

Property	Value
SMILES	`CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`
True Label	1 (Active)
Predicted Label	1 (Active)
Confidence	100%

Now let me run all four explainability methods on this compound:

(Calling tool `explain_with_molanchor`)
(Calling tool `explain_smiles_with_shap`)
(Calling tool `explain_with_molce`)
(Calling tool `generate_counterfactuals`)

Assistant

🔬 All four XAI methods completed! Let me get the SHAP visualization:

(Calling tool `plot_shap_mol`)

Assistant

Now let me visualize the MolCE results and the counterfactuals:

(Calling tool `visualize_counterfactuals`)

Assistant

🎉 **Complete XAI Analysis Finished!** Here's the comprehensive summary:

📊 Model Performance Summary

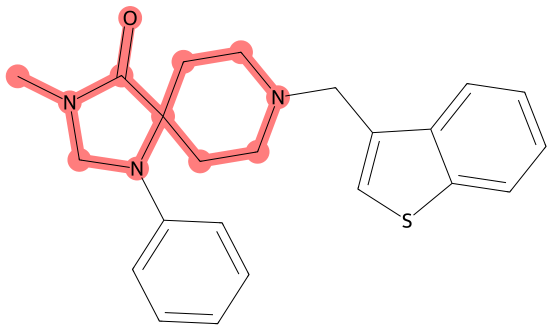
Metric	Value
Test Accuracy	98.64%
AUC-ROC	0.9957
Balanced Accuracy	0.9864
F1 Score	0.9864
Precision	0.9849
Recall	0.9880

🔬 XAI Analysis on Selected Active Compound

Compound: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`

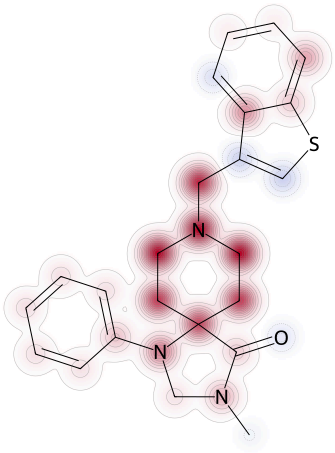
📍 MolAnchor Results (Fragment Anchors)

Anchor SMARTS	Precision	Coverage	Interpretation
`[*]N1CCC2(CC1)C(=O)N(C)CN2[*]`	100%	53%	**Morpholinone-piperazine core** — highly precise anchor present in 53% of actives



📈 SHAP Analysis (Feature Contributions)

Metric	Value
SHAP Sum	53.69 (strong positive)
Prediction	Class 1 (Active)
Key Insight	Core amide/heterocycle fragments show strongest positive contributions



🔄 MolCE Results (Contrastive Attribution)

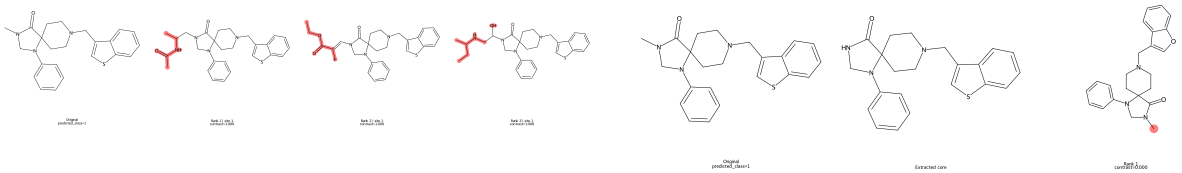
R-Group Analysis (Top 3 Contrastive):

Rank	R-Group SMILES	Site	Contrast Score	Effect
1	`*CC(C)NC(C)=O`	1	**1.0**	Replacing would shift toward inactive
2	`*C=C(F)C(=O)OCC`	1	**1.0**	Replacing would shift toward inactive
3	`*C(O)CNC(C)CC`	1	**1.0**	Replacing would shift toward inactive

Scaffold Analysis:

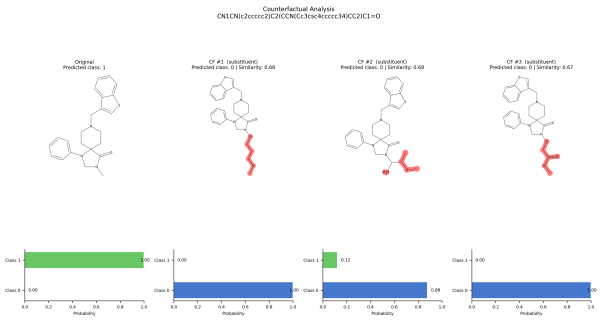
Rank	Scaffold SMILES	Contrast Score
1	`O=C1N([*:1])CN(c2ccccc2)C12CCN(Cc1coc3ccccc13)CC2`	**0.0**

Key Finding: Scaffold has **zero contrast score** — the core structure is stable and essential.



🎯 Counterfactual Analysis (What Makes it Inactive?)

Rank	Counterfactual SMILES	Predicted Class	Probability	Tanimoto	Change Type
1	`COCCCON1CN...`	**0 (Inactive)**	99.98%	0.68	+ethoxyethoxy chain
2	`COC(C)C(N)N1CN...`	**0 (Inactive)**	87.76%	0.68	+isopropylamine
3	`COC(=O)CON1CN...`	**0 (Inactive)**	100%	0.67	+ester chain



📋 Summary: Agreements & Disagreements Across Methods

✅ AGREEMENTS (All Methods Agree)

Aspect	Consensus Finding
Core Importance	✅ All methods agree the **morpholinone-piperazine core** is critical for activity
Core Stability	✅ MolCE scaffold score = 0.0 + MolAnchor high coverage = core is essential
Substituent Sensitivity	✅ Counterfactuals + MolCE R-groups agree that **changing substituents flips prediction**
Activity Driver	✅ SHAP positive contributions + MolAnchor anchors point to the same core structure

⚠️ DIFFERENCES (Complementary Perspectives)

Aspect	MolAnchor	SHAP	MolCE	Counterfactuals
Granularity	Fragment-level (SMARTS patterns)	Atom-level (per-atom contributions)	Scaffold/R-group level	Molecular level (full structures)
Focus	"What fragments are anchors?"	"Which atoms contribute most?"	"Why active vs inactive?"	"What changes make it inactive?"
Output Type	Precise SMARTS patterns	Quantitative atom scores	Contrastive scores	Alternative molecules
Key Insight	Core = anchor (100% precision)	Core atoms = highest SHAP	Scaffold = neutral (0.0)	Substituents = flip point

🔍 Key Takeaways

1. **Consensus:** The **morpholinone-piperazine core** (`[*]N1CCC2(CC1)C(=O)N(C)CN2[*]`) is the critical pharmacophore for D_2 receptor activity
2. **Substituent Effect:** Adding polar/extended chains (ethoxy, ester, amine) to the core **flips prediction to inactive**
3. **Scaffold Stability:** The core scaffold itself has **zero contrast score** — it's the stable, essential feature
4. **Complementary Views:**
   - MolAnchor identifies the **anchor fragment** precisely
   - SHAP shows **which atoms** within that fragment matter most
   - MolCE explains **why it's active vs inactive** at R-group level
   - Counterfactuals show **what specific changes** would destroy activity

🧪 Chemical Interpretation

The D_2 dopamine receptor active compounds share a **morpholinone-piperazine core** that likely interacts with the receptor's binding pocket through:

- **Hydrogen bonding** via the amide carbonyl
- **π-π stacking** via the phenyl rings
- **Hydrophobic interactions** via the benzothiazole moiety

Substituent modification (adding polar chains like ethoxy or ester groups) disrupts this interaction pattern, flipping the prediction to inactive.